

PPG-GAN: An Adversarial Network to De-noise PPG Signals during Physical Activity

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Abstract—Quality photoplethysmographic (PPG) signals are essential for accurate physiological assessment. However, the PPG acquisition process is often accompanied by spurious motion artefacts (MAs), especially during medium-high intensity physical activity. This study proposes a generative adversarial network (PPG-GAN) to create de-noised versions of measure PPG signals. The Adaptive Notch Filtration (ANF) algorithm, which enables the extraction of accurate heart rates (HR) and respiration rates (RR) from PPG signals, is used as the approximate reference signal to train the PPG-GAN. The generated PPG signals from test inputs provide a heart rate (HR) with a mean absolute error of 1.68 bpm for the IEEE-SPC dataset. A comparison with gold-standard HR and RR measurements, for our in-house dataset, show the errors in absolute value of less than 5%. The generated PPG signals, for the test clips, show a very strong correlation with their reference values, $R \approx 0.98$. The results suggest that PPG-GAN could be a paradigm for MA-free PPG signal processing specifically for personal healthcare, even during high intensity activity.

Index Terms—photoplethysmography, generative adversarial network, motion artefacts, heart rate, respiration rate

I. INTRODUCTION

The pulsatile waveform, most commonly extracted through electrocardiography (ECG) or photoplethysmography (PPG), is a useful physiological signal in monitoring vital signs, including the heart rate (HR) and the respiration rate (RR). ECG requires multiple electrodes, at different locations on the body making it a rather inconvenient technique, especially during exercise, and unsuitable for integration into a wearable device. By contrast, PPG is a convenient optical technique for non-invasive detection of the volumetric blood changes based on the light passing through or reflected from the skin tissue [1]. Obtaining stable physiological parameter readings is dependent on the quality of the PPG signal; the clearer the features of the PPG signal, the higher their accuracy. Motion artefacts (MA), created during physical activity, are a hindrance to the acquisition of clear PPG signals, reducing the stability and accuracy of the monitored vital signs. These MA may result from motion-induced changes in the contact between the wearable sensor and the skin surface, as well as haemodynamic effects [2]. Several methods have been presented with a view to reducing MA and extracting cleaned PPG signals. These include Empirical Mode Decomposition (EMD), Fast Fourier Transformation (FFT), Independent Com-

ponent Analysis (ICA), and wavelet denoising approaches [3]. Perhaps the most common is some form of Adaptive Noise Cancellation (ANC) using Recursive Least Squares (RLS) or Adaptive-Size Least Mean Squares (AS-LMS) [4]. These methods tend to perform poorly during high intensity physical activities, an issue addressed by TROIKA [5]. However, this MA-removal method is complex and challenging to implement efficiently in a wearable system [6]. Consequently, it is important to develop other PPG algorithms, which both perform well under all conditions but which are also appropriate for wearable systems. Such a prospect seems likely to lie in the domain of Artificial Intelligence.

Deep learning (DL) algorithms for the extraction of physiological parameters from PPG signals have been presented in recent years. These methods can generally be divided into the *end-to-end* type and *feature extraction* type. The former directly establishes the mapping from PPG signals to the target physiological parameter [7], while the latter get the physiological parameter targets by extracting the feature information pre-processed from PPG signals [8]. Since DL is data-driven, and neural networks have strong fitting abilities, the results of DL-based PPG methods often outperform traditional methods [9]. However an ability to also perform well during high intensity activity creates a specific additional problem.

To calculate a Heart Rate from a PPG signal free of motion noise, the network essentially needs to learn how to select the dominant frequency from some sort of spectral analysis. In the presence of motion noise, the level of difficulty presented to the network is increased, as may be seen in the ten-second clip shown in Fig. 1, from a subject exercising on a treadmill, during relatively high-intensity activity ($HR \approx 170\text{bpm}$). A optoelectronic sensor, attached to the subject's thumb, also provides triaxial acceleration readings and the cardiac signal was simultaneously monitored with an ECG probe for verification purposes.

In the clip in Fig. 1, the spectrum of the x-axis accelerometer signal ($\hat{A}_x(f)$, bottom right) has peaks at around 1.31 Hz or 78.60 cycles per minute, and 2.63 Hz or 157.80 cycles per minute. These frequencies find their way into the measured signal ($x(t)$, whose spectrum $\hat{X}(f)$ is top right) most probably as a result of a cyclic variation in contact pressure caused by the motion generated through the exercise and the impact

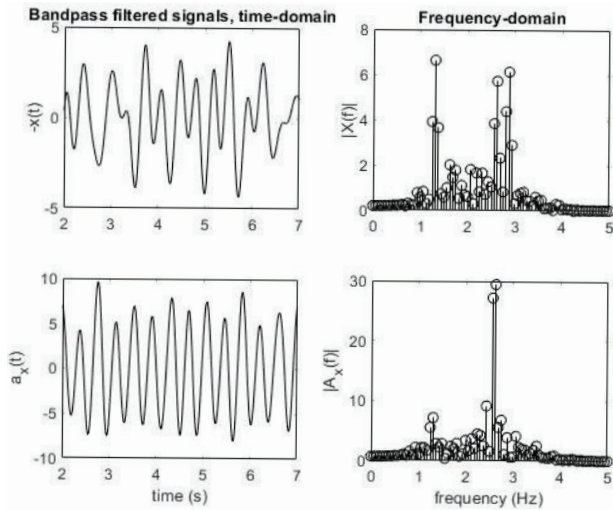


Fig. 1. The top row shows time-domain and frequency-domain representations of a 10s clip of the measured PPG signal (after standard bandpass filtering of between 0.5 and 6.0 Hz). The second row contains similar results for the x -axis of the accelerometer signal (the y and z axes are similar). For clarity only the central five seconds of the time-domain signals are shown.

of the motion $a_x(t)$ on the measured signal $x(t)$ seems fairly clear in the time-domain figures (top and bottom left, respectively of Fig. 1). If the network were to behave only as a maximum amplitude selector of the spectrum, the wrong heart rate of around 78.75 bpm would be returned, rather than the correct value, verified by ECG, of around 172.5 bpm which corresponds to the second highest peak in $\tilde{X}(f)$. Nevertheless CorNET [8], a Convolutional Neural Network (CNN), has successfully extracted a subject's HR, from a measured PPG signal, in the presence of motion noise generated through an exercise protocol at a variety of intensities. This suggests that, during training, the CorNET network has learned to recognise the difference between a motion-induced peak and a heart rate peak, and to ignore those related to motion.

This success raises the question, which we consider in this paper, of whether, with the right network, this skill might be extended from not only HR but also to include the full pulsatile waveform. In that case, what would be required of the network, during its inference phase, is the ability to take a noisy signal (the measured PPG signal $x(t)$, with MAs included) as its input, and to output a clean signal $p(t)$ (free of such artefacts).

A. Generative Adversarial Networks

The extraction of the desired PPG signals $p(t)$ from the measured signal $x(t)$ can set up as a generative adversarial problem and so should be amenable to network analysis. A generative adversarial network (GAN), first proposed by Goodfellow in 2014 [10], is comprised of two networks, a generator (G) and a discriminator (D) which are trained in an adversarial (minimax) way; G generates *fake* signals that can confuse D , while D judges whether the generated signals $p(t) = G(x(t))$ is a *fake* or a *true* one. Recent works with GANs in this, and related areas, are quite promising:

in [11], a GAN was used to remove low intensity motion noise from PPG signals by representing the PPG as a two dimensional (correlation) image array allowing a standard image-based GAN architecture to be used; and the removal of ocular artifacts from electroencephalography (EEG) data, necessary in a variety of brain-computer interface applications, was attempted in [16].

A general issue, with a method such as this, is that noise-free data is generally needed for training purposes. In [11], where the motion effects are only due to sensor motion (and not complicated by the change in the pulsatile cardiac signal necessary to create that motion, such as occurs in medium-high intensity exercise), the periods where the sensor is at rest may be used as noise-free (ground truth) data. Conversely, in [16], state-of-the-art electro-oculography based algorithms are assumed to provide the best approximation to cleaned data, but it will still only be an approximation. In the present case, involving several intensities of physical activity, a three-axis accelerometer (and perhaps other sensor outputs) provides an equivalent information source to that of the electro-oculography data used in [16]. State-of-the-art motion removal algorithms can be used here as a ground truth reference and as the state-of-the-art improves it may be expected that the performance of an appropriately trained GAN will also improve.

B. Focus of Current Work

The methodology for using GANs for real-time removal of MAs from the PPG signal is essentially to :

- Obtain a set of measured pulsatile waveforms using a PPG device throughout a defined exercise protocol for use in network training and testing.
- Divide the set of waveforms into short (8-10 s) clips, and split between a training set ($x(t) \in \mathcal{X}_{Tr}$) and a testing set ($x(t) \in \mathcal{X}_{Te}$).
- Then, offline, use a state-of-the-art motion removal algorithm to create paired reference data $p_{ref}(t)$, for the training/testing data.
- Train a GAN network using the paired training data clips ($x(t), p_{ref}(t)$), for $x(t) \in \mathcal{X}_{Tr}$ and test the result by comparing the generator output $p(t) = G(x(t))$ against the known reference $p_{ref}(t)$ for all $x(t) \in \mathcal{X}_{Te}$. Install on the real-time device.
- Perform real-time cleansing of a random $x(t)$ to create a clean version $p(t) = G(x(t))$ using the trained generator.

The advantage of using the GAN, in the removal of MAs, is that computationally heavy state-of-the-art signal processing is only necessary during the offline training phase and not during in-use inference.

With this in mind, the aim of the current study is to test the principal that a GAN can be used to clean noisy signals in an effective manner. Here, we use a simplified version of the cleansing process, the ANF algorithm, which is able to accurately extract both HR and RR, including during medium-high intensity physical activity [12]. We thus solve a proxy problem which replaces the waveforms $p(t)$ which

would nominally be created using the state-of-the-art cleaning methods with the ANF algorithm to create ANF-clean, paired data $p_{ref}(t) = A(x(t))$ from measured training/testing data $x(t) \in \mathcal{X}_{tr} \cup \mathcal{X}_{te}$.

II. METHODOLOGY

The proposed PPG-GAN framework is depicted in Fig. 2. It consists of three main parts: 1) Generator, 2) Discriminator, and 3) ANF. During the training, the generator takes a MA-corrupted PPG signal $x(t) \in \mathcal{X}_{Tr}$ as an input and generates an approximation to the desired PPG signal $p(t) = G(x(t))$ as its output. The discriminator is designed to compare the scalars $D(p(t); x(t))$ with $D(p_{ref}(t); x(t))$, giving a probability (between 1 and 0) that $G(x(t))$ coming from the reference training data. Finally, ANF algorithm extracts the reference PPG signals $p_{ref}(t) = A(x(t))$ for the training purpose. To testify the quality of PPG-GAN waveforms, the tested signals $p(t) = G(x(t))$ for $x(t) \in \mathcal{X}_{Te}$ are compared to the ANF results $p_{ref}(t)$, and the physiological parameters (HR and RR) are extracted from the de-noised PPG signals.

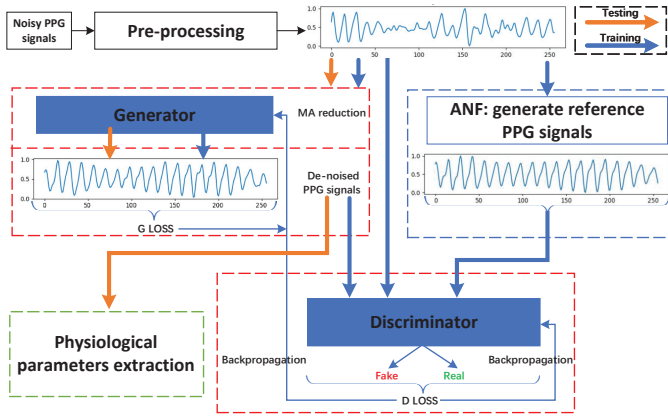


Fig. 2. The overall framework of the PPG-GAN network.

A. Acquisition of Reference PPG signals

As indicated above, the ANF algorithm is used to obtain the reference PPG signals $p_{ref}(t)$. The ANF algorithm is capable of extracting an approximate MA-free PPG signals and accurate physiological parameters. It is based on adaptive moving average filter and notch filters, to remove out-of-band and in-band noise, and works at different motion intensities. Specifically, PPG signals and three-axis acceleration signals are inputs to the ANF and the reference signals are extracted by finding a set of peaks from the accelerometer frequency spectrum and applying the adaptive notch filter to filter out peaks considered to arise from motion noise.

B. 1D PPG-GAN Framework

To better pair $x(t)$ and $p(t) = G(x(t))$, the measured signal $x(t)$ is set as a second (conditional) input to discriminator. Hence, the input from discriminator comes from the three signals, $x(t)$, $p(t) = G(x(t))$, and $p_{ref}(t)$. The output from discriminator is thus conditioned on $x(t)$. Once

trained discriminator should not be able to distinguish between $D(p_{ref}(t); x(t))$ and $D(G(x(t)); x(t))$ for the training samples. The network structure uses a U-net network, similar to pix2pix reported in [13] where further details may be found. Fig. 3 shows the generator as a de-noising encoder-decoder with some skip connections [14]. Low-level detail can be lost if all feature information is forced through the compression bottleneck at the centre of the network [15]; the idea of the skip connections being to transfer fine-grained features from the encoding stage to the decoding stage directly.

1D convolution and deconvolution functions are adopted to replace the 2D functions in pix2pix, and the latent vector z is also removed. These modifications are more suitable for 1D signal processing and reduce the overfitting in generating desired PPG signals. In detail, the left-hand side is similar to an encoding stage, containing eight convolutional layers followed by batch normalization (BN) and leaky rectified linear (LeakyReLU) units. The decoding process is the reversal of the encoding process. The generator network also applies skip connections, connecting each encoding layer to the corresponding decoding layer and bypassing the compression process in the middle of generator network. Specifically, the input and output of generator network share the same underlying structure, which is that of measure PPG test/training samples. Discriminator network is structured in five 1D-convolutional

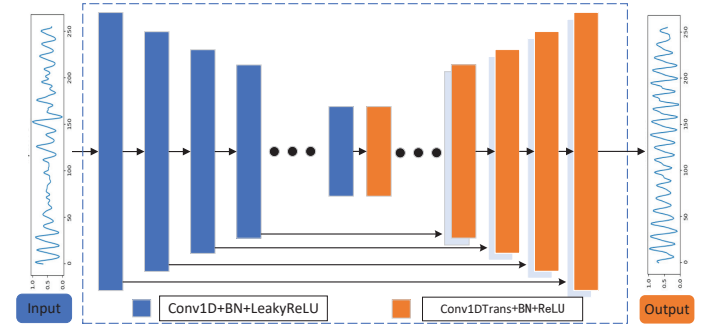


Fig. 3. The generator network based on the pix2pix network [13].

layers with a fully connected final layer, as illustrated in Fig. 4. The leaky rectified linear (LeakyReLU) is selected for the nonlinear activation function, and batch normalization is adopted to speed up the convergence. Discriminator uses (conditional) probabilities to judge the two sets of signal pairs, the generated probability $D(G(x(t)); x(t))$ and the reference probability $D(p_{ref}(t); x(t))$.

C. Loss Function

The noise removal problem of the PPG signal can be presented as $x(t) = p(t) + n(t)$, $t \geq 0$ where $x(t)$ is the raw measured PPG signal, including the MAs, $p(t)$ is the desired PPG signal, and $n(t)$ is the noise to be removed. In the discrete case

$$p[i] = x[i] - n[i], i = 1, 2, 3 \dots \quad (1)$$

where i is the sampling index, and the $\mathbf{x} = [x[i]]$, $\mathbf{n} = [n[i]]$ and $\mathbf{p} = [p[i]]$ are all 1D vectors. Using its strong ability

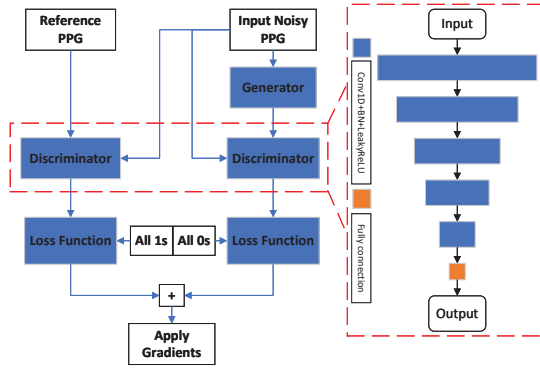


Fig. 4. Discriminator network based on conv1D/BN/Leaky-ReLU layers.

to learn the difference between the noisy PPG signal and the reference PPG signal, the PPG-GAN can learn the distribution features of the noisy PPG signals $\mathbf{x}$ and of the reference signals $\mathbf{p}_{ref}$ thus enable to generate the de-noised PPG signal $G(\mathbf{x})$. The *PPG feature loss*, the error between $\mathbf{p}_{ref}$ and $G(\mathbf{x})$ in the time domain, is given by

$$L_{ppg} = \|G(\mathbf{x}) - \mathbf{p}_{ref}\|_1 \quad (2)$$

where $\|\cdot\|_1$ represents the $L1$ norm. This error is added to the binary cross entropy loss, with weighting λ , to give a total *generator loss function* of

$$L_G = \log(1 - D(G(\mathbf{x}); \mathbf{x})) + \lambda L_{ppg} \quad (3)$$

The *discriminator binary cross entropy loss function* is [15]

$$L_D = \log(D(\mathbf{p}_{ref}; \mathbf{x})) + \log(1 - D(G(\mathbf{x}); \mathbf{x})) \quad (4)$$

with which discriminator differentiates between the generated and reference PPG signals. In training, the generator minimizes the error losses against the adversarial discriminator which tries to maximize error losses. In this way the generator continuously learns global and local features, and the generated PPG signal maintains the characteristics of the reference signal as closely as possible.

III. EXPERIMENTS

A. Experimental Setup

An eight second sliding window is used to process all input and reference PPG signals into the clips $x(t)$, with a sliding step set to one second. All input and reference PPG signals are downsampled to 32 Hz. In experiments, the proposed framework is run for 200 epochs, and the generator and discriminator networks are simultaneously optimized using the Adam optimizer with the learning rate of 0.0002. The weight parameter λ in (3) is set to 100.0, to balance the PPG waveform loss. The input PPG data is pre-processed with a fourth order Butterworth band-pass filter with band edges set at 0.2 Hz and 6 Hz.

B. Measurement Protocol

The exercise datasets used here are provided by Loughborough University (the in-house dataset) and the IEEE Signal Processing Cup 2015 (the IEEE-SPC dataset). The in-house dataset was recorded using a four-channel multi-wavelength illumination optoelectronic patch sensor (mOEPS) at a sampling frequency of 256Hz, from twelve subjects with ages from 23 to 37, while exercising on a treadmill [12]. Subjects performed the exercise at four different intensities (3kmh^{-1} , 6kmh^{-1} , 9kmh^{-1} , 12kmh^{-1}); each stage lasting four minutes with a recovery of 60s between. A reference HR was recorded using a Polar Bluetooth® Smart chest strap (Polar, Electro, Kempele, Finland) and a reference RR was obtained from a Vyntus™ CPX Metabolic cart (JAEGER™ Vyntus™ CPX, Carefusion, Germany).

The IEEE-SPC datasets consist of a two-channel PPG signal with green LEDs (wavelength = 515nm), triaxial accelerometer signal, and an ECG reference signal, for twelve male subjects ages 18–35, at a sampling frequency of 125Hz [5].

C. Experimental Results

To assess the success of the method, HR and RR are calculated separately for the generated signals $G(x(t))$ and the reference PPG signals $p_{ref}(t)$, and both were compared with those from the commercial devices for the in-house dataset. We extracted a total of 15366 samples, which are divided into the training set and testing set with a ratio of 0.8 and 0.2. The testing results of continuous HR and RR monitoring in the treadmill exercise are presented in Fig. 5 and 6, where they are compared with the two different references. *Polar* in Fig. 5 is a reference heart rate recorded using a Polar Bluetooth Smart chest strap and the *Reference* RR is collected by a Vyntus™ CPX Metabolic cart. In both figures the ANF reference uses our previously published algorithm [12]. The performance in the extraction of HR and RR is robust and highly correlated with the two references provided. As the fluctuations in a continuously measured RR are quite significant (Fig. 6), the blue shaded part is included to represent the average value of the RR in the different exercise stages and represents the change in respiration rate more intuitively. Fig. 7 presents detailed PPG waveform in the time domain from the different exercise intensity points shown in Fig. 5, to further illustrate the quality of the generated PPG signals. Points A, B, C and D are selected at exercise intensities of 3 kmh^{-1} , 6 kmh^{-1} , 9 kmh^{-1} and 12 kmh^{-1} . The generated PPG signals show that the proposed framework can produce satisfactory results that are close to the reference signals at different exercise intensities. A measure of the quality of the proposed PPG-GAN framework in removing MA noise is the correlation R between the generated PPG signal $G(x(t))$ and the ANF reference $p_{ref}(t) = A(x(t))$, for each vector $x(t)$ in the test set, $\mathcal{X}_{Te}$.

$$R = \frac{\int_0^T G(x(t))p_{ref}(t)dt}{\sqrt{\int_0^T G(x(t))^2dt}\sqrt{\int_0^T p_{ref}(t)^2dt}} \quad (5)$$

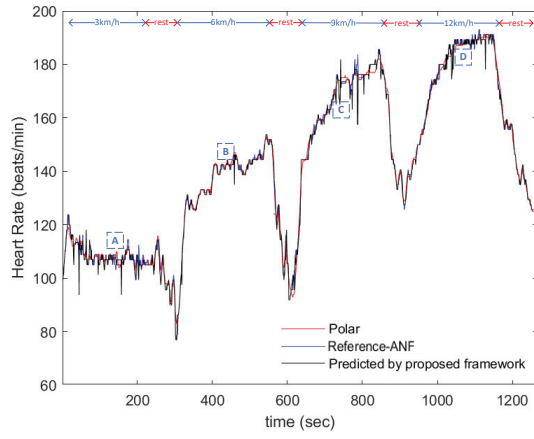


Fig. 5. HR calculation results on M002 subject chosen in-house dataset during different exercise intensities.

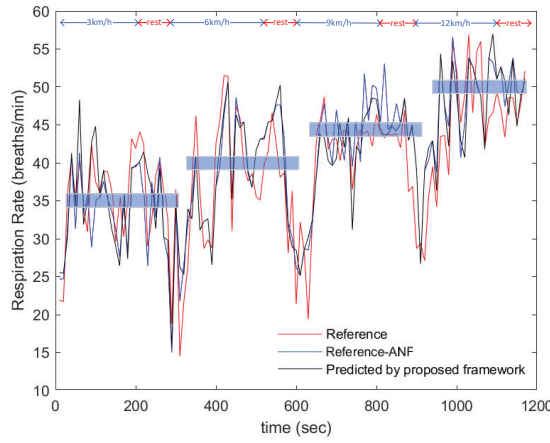


Fig. 6. RR calculation results on M002 subject chosen in-house dataset during different exercise intensities.

where T is the length of each of the test clips (i.e., $T=8s$). Fig. 8 shows that the PPG-GAN outputs are very highly correlated with the ANF reference signals across the entire set of test vectors for this subject. The mean value $\langle R \rangle = 0.98$ and the standard deviation $\langle (R - \langle R \rangle)^2 \rangle = 0.01$.

The RR performance of the proposed framework is also investigated in Fig. 9 as a Bland-Altman plot to illustrate the consistency of the calculated RRs from PPG-GAN, ANF and the gold-standard RRs from the VyntusTM CPX Metabolic cart device. The limit of agreement (LOA) for PPG-GAN is $[-8.91, 10.63]$ breaths/min, and $[-8.58, 10.43]$ breaths/min for ANF. Furthermore, 95% of all differences lie inside this range. The performance for HR calculations is better still with a limit of agreement (LOA) $[-5.02, 4.38]$ bpm.

To evaluate the PPG-GAN performance, the IEEE SPC dataset was selected as a benchmark, which comprises twelve training datasets and ten testing datasets, as shown in Table I. The mean absolute error (Err_1) and the standard deviation (SD) are computed and compared with other reported methods using a 10-subject testing dataset. This table shows that, for the severely noisy ten test subjects, the proposed framework

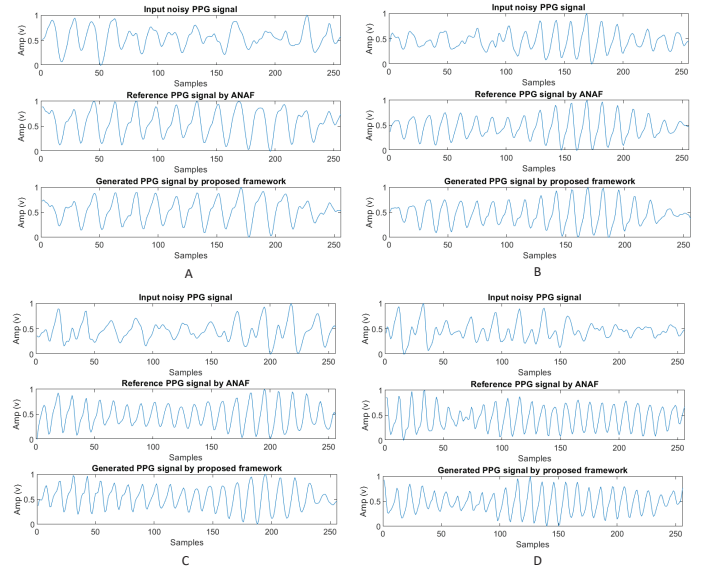


Fig. 7. A Comparison of the original noisy signal, the reference signal and the generated signal at the different exercise intensities defined in Fig. 5.

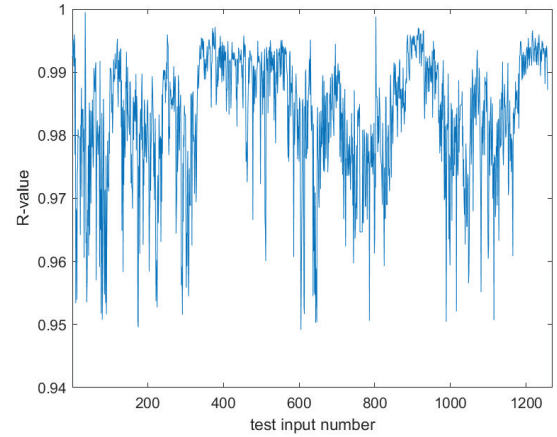


Fig. 8. The coefficient of correlation R between the generated PPG signal and the ANF ground truth on M002 subject.

gave HR calculations with Err_1 of 1.68 and SD of 0.76 beats/min. From the Err_1 values, the proposed framework has a better performance than the other traditional de-noised methods, i.e., TROIKA [5] and MURAD [4], but is not yet as good as the CorNET [8] framework that adopts the deep learning architecture.

IV. DISCUSSION AND CONCLUSION

The experimental results show that the PPG-GAN has the effectiveness to target PPG signal extraction from high intensity motion noise. Specifically, the test results of in-house and IEEE-SPC datasets show a significant benefit to the accuracy when using adversarial and time-domain losses to account for both global and local features of PPG signals.

The use of an in-house dataset has allowed reference readings of HR and RR to be gathered as the benchmark to help assess the 1D PPG-GAN framework process for accurate

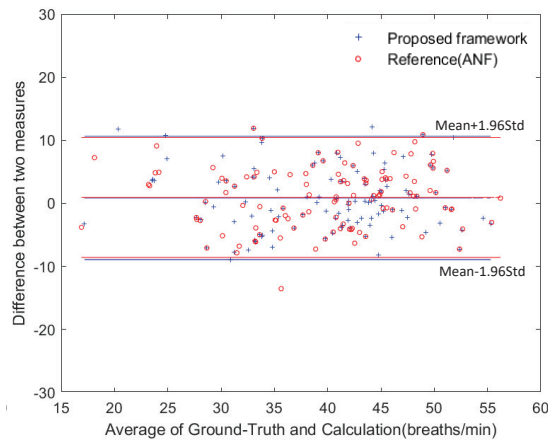


Fig. 9. Bland-Altman plot of between calculated RR and the polar RR on in-house dataset: PPG-GAN vs ANF method.

TABLE I
COMPARISON OF Err_1 (IN BEATS/MIN) ON THE IEEE-SPC OF THE PROPOSED FRAMEWORK WITH THE SAME OF OTHER RECENTLY REPORTED RESULTS.

Subject	TROIKA	MURAD	CorNET	PPG-GAN
1	6.63	6.35	1.60	2.77
2	1.94	1.15	0.24	1.05
3	1.35	1.51	1.60	1.53
4	7.82	2.96	2.04	2.75
5	2.46	2.78	0.95	2.25
6	1.73	1.54	0.28	1.44
7	3.33	2.03	0.28	1.96
8	3.41	2.02	0.67	1.77
9	2.69	0.97	0.42	0.74
10	0.51	0.53	0.57	0.60
Mean	3.18	2.18	0.86	1.68
(SD)	(2.32)	(1.64)	(0.65)	(0.76)

extraction of HR and RR readings. The IEEE-SPC datasets were used as a benchmark to compare the performance of PPG-GAN with other recently reported algorithms. In Table 1, it can be seen that the proposed framework has better HR extraction accuracy, for the IEEE-SPC datasets, than other traditional algorithms. While the CorNET framework has higher HR extraction accuracy (0.86 bpm < 1.68 bpm) compared to the proposed framework, it so far only assesses the accuracy of a single physiological parameter, i.e., HR. By contrast, the PPG-GAN aims to provide full pulsatile waveform with the MAs removed as anticipated, allowing multiple physiological parameters to be obtained.

In this study, the 1D PPG-GAN framework has been created to extract quality PPG signals for physiological monitoring and assessment. PPG-GAN is a generative adversarial network (GAN) allow a MA-corrupted PPG signal as its input, and to output a cleansed signal. The quality of the generated output is tested through its correlation with reference signals and HR and RR assessment. The experimental results on the in-house

dataset and the IEEE-SPC dataset show that the PPG-GAN removes MAs effectively and enhances the quality of noisy input PPG signals. The motion artefact-free PPG signal can be used to determine HR and RR with an accuracy of above 95% against gold standard references.

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